

# Coding for Final Project

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## Data Cleaning

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
# importing datasets
suns = pd.read_csv("DataFolder/Suns_21-25.csv")
kings = pd.read_csv("DataFolder/Kings_21-25.csv")
mavs = pd.read_csv("DataFolder/Mavs_21-25.csv")
spurs = pd.read_csv("DataFolder/Spurs_21-25.csv")

suns["Team"] = "Suns"
kings["Team"] = "Kings"
mavs["Team"] = "Mavs"
spurs["Team"] = "Spurs"

# putting them all into one dataset called 'games'
games = pd.concat([suns, kings, mavs, spurs], ignore_index=True)
```

```
# renaming columns
games = games.rename(columns={"Unnamed: 3": "HomeAway", "Opp.1": "OppPoints"})

# Adding variables Win, Home game, point difference, 2 point attempt rate
games["Win"] = games["Rslt"].map({"W": 1, "L": 0})
games["Home"] = np.where(games["HomeAway"] == "@", 0, 1)
games["PointDiff"] = games["Tm"] - games["OppPoints"]
games["2PAr"] = games["2PA"] / games["FGA"]
```

```

# Remove extra columns
drop_cols = ["Rk", "Gtm", "OT", "eFG%.1", "eFG%.2", "eFG%.3", "TOV%.1", "ORB%.1", "FT/FGA.1"]
games = games.drop(columns=drop_cols, errors="ignore")

# Change Date col format & create a Season col
games["Date"] = pd.to_datetime(games["Date"], format="mixed", errors="coerce")
games["Season"] = games["Date"].dt.year

# Make columns numeric
num_cols = ["Tm", "OppPoints", "PointDiff", "ORtg", "DRtg", "Pace", "3PAr", "2PAr", "TOV%", ""]
for col in num_cols:
    games[col] = pd.to_numeric(games[col], errors="coerce")

games.head()

```

|   | Date       | HomeAway | Opp | Rslt | Tm  | OppPoints | FG | FGA | FG%   | 3P | ... | BLK% | TOV% |
|---|------------|----------|-----|------|-----|-----------|----|-----|-------|----|-----|------|------|
| 0 | 2021-10-20 | NaN      | DEN | L    | 98  | 110       | 36 | 87  | 0.414 | 14 | ... | 6.8  | 16.0 |
| 1 | 2021-10-22 | @        | LAL | W    | 115 | 105       | 42 | 86  | 0.488 | 10 | ... | 3.8  | 10.9 |
| 2 | 2021-10-23 | @        | POR | L    | 105 | 134       | 43 | 89  | 0.483 | 9  | ... | 2.3  | 12.8 |
| 3 | 2021-10-27 | NaN      | SAC | L    | 107 | 110       | 42 | 93  | 0.452 | 6  | ... | 10.5 | 9.6  |
| 4 | 2021-10-30 | NaN      | CLE | W    | 101 | 92        | 39 | 84  | 0.464 | 6  | ... | 6.4  | 13.5 |

## EDA

```

# Create df just for the EDA
eda_df = games[["Team", "Season", "Date", "Opp", "Home", "Win", "Tm", "OppPoints", "PointDiff", ""]]

# Summary of team data
team_summary = eda_df.groupby("Team")[["Win", "PointDiff", "ORtg", "DRtg", "Pace", "3PAr", "2PAr", "TOV%", ""]]
print(team_summary)

```

|       | Win      | PointDiff | ORtg       | DRtg       | Pace       | 3PAr     | \ |
|-------|----------|-----------|------------|------------|------------|----------|---|
| Team  |          |           |            |            |            |          |   |
| Kings | 0.500000 | -0.149390 | 115.774085 | 115.992988 | 99.228659  | 0.406479 |   |
| Mavs  | 0.545732 | 1.097561  | 115.612805 | 114.483841 | 97.773476  | 0.439073 |   |
| Spurs | 0.341463 | -4.789634 | 111.874085 | 116.563720 | 100.525610 | 0.384439 |   |
| Suns  | 0.591463 | 2.408537  | 115.998777 | 113.647095 | 98.308563  | 0.383893 |   |

|       | 2PAr     | TOV%      | ORB%      | TS%      | eFG%     |
|-------|----------|-----------|-----------|----------|----------|
| Team  |          |           |           |          |          |
| Kings | 0.593564 | 11.982927 | 23.304573 | 0.584329 | 0.550915 |
| Mavs  | 0.560972 | 11.686280 | 21.210976 | 0.587448 | 0.555043 |
| Spurs | 0.615617 | 12.418293 | 24.088720 | 0.563558 | 0.533555 |
| Suns  | 0.616387 | 12.471865 | 23.855352 | 0.588771 | 0.553814 |

- Suns have best results overall
  - highest win rate: 0.591
  - best point difference: +2.409
  - best defensive rating: 113.647
- Suns & Mavs similar offenses
- Defense is biggest separator
  - Suns DRtg: 113.647 best in group
  - Mavs DRtg: 114.484
  - Kings DRtg: 115.993
  - Spurs DRtg: 116.564
- Mavs have the highest 3PAr: 0.439; Suns are not winning bc they shoot the most 3s - they have more variety in shots and better defense
- Suns have highest TOV%

```
# Suns Wins & Losses table
suns_df = eda_df[eda_df["Team"] == "Suns"].copy()
suns_wins_losses = suns_df.groupby("Win")[["ORtg", "DRtg", "3PAr", "2PAr", "TOV%", "ORB%", "TS%", "eFG%", "PointDiff"]]
print(suns_wins_losses)
```

|     | ORtg       | DRtg       | 3PAr     | 2PAr     | TOV%      | ORB%      | \ |
|-----|------------|------------|----------|----------|-----------|-----------|---|
| Win |            |            |          |          |           |           |   |
| 0   | 111.892537 | 120.396269 | 0.391843 | 0.612910 | 13.224627 | 23.632090 |   |
| 1   | 118.849741 | 108.961140 | 0.378373 | 0.618788 | 11.949223 | 24.010363 |   |

|     | TS%      | eFG%     | PointDiff  |
|-----|----------|----------|------------|
| Win |          |          |            |
| 0   | 0.571201 | 0.526933 | -11.477612 |
| 1   | 0.600969 | 0.572381 | 12.000000  |

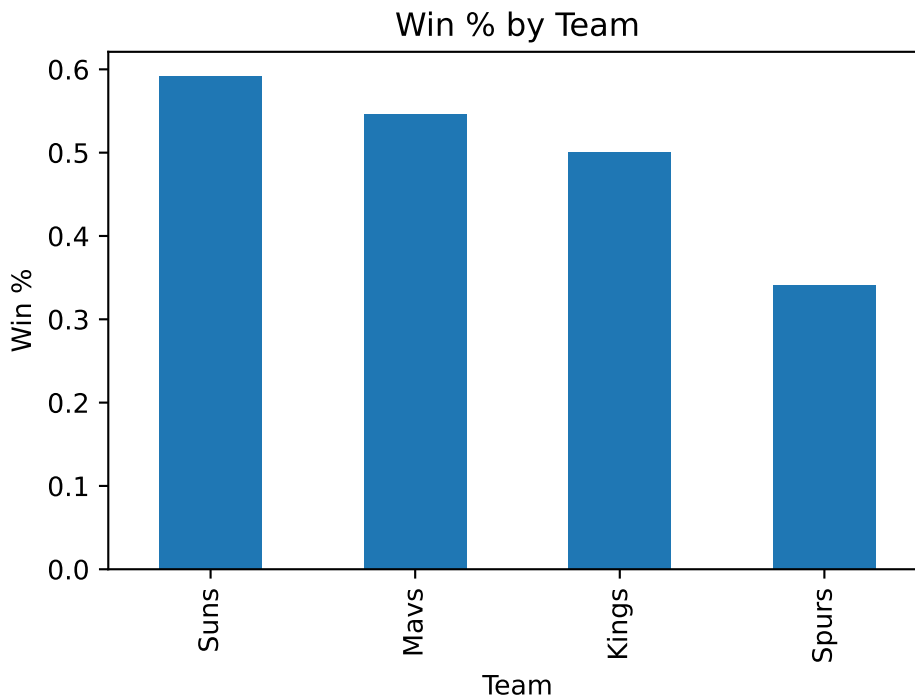
When Suns win, they have better defense, better shooting accuracy, less turnovers. Suns success is more based on efficiency and defense than shot selection & three pointers.

## Team Comparison

```
# Win Percentage by Team bar chart
plt.figure()
win_pct = eda_df.groupby("Team")["Win"].mean().sort_values(ascending = False)
plt.figure()
win_pct.plot(kind = "bar")
plt.ylabel("Win %")
plt.title("Win % by Team")
```

Text(0.5, 1.0, 'Win % by Team')

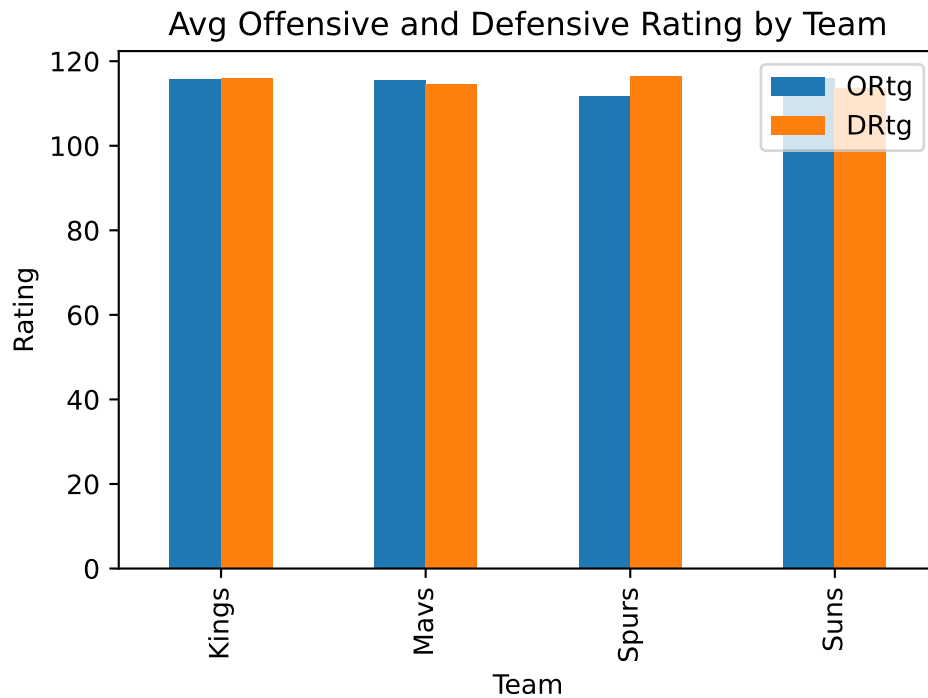
<Figure size 1650x1050 with 0 Axes>



```
# ORtg and DRtg by team
plt.figure()
ratings = eda_df.groupby("Team")[["ORtg", "DRtg"]].mean()
ratings.plot(kind = "bar")
plt.title("Avg Offensive and Defensive Rating by Team")
plt.ylabel("Rating")
```

```
Text(0, 0.5, 'Rating')
```

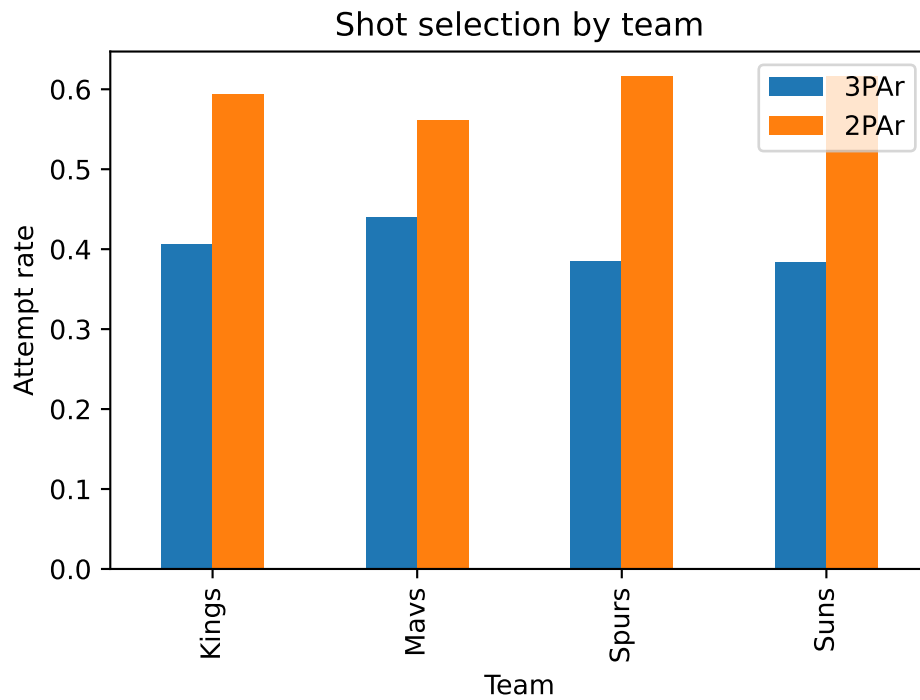
```
<Figure size 1650x1050 with 0 Axes>
```



```
# Shot type by team
plt.figure()
shot_type = eda_df.groupby("Team")[["3PAr", "2PAr"]].mean()
shot_type.plot(kind = "bar")
plt.title("Shot selection by team")
plt.ylabel("Attempt rate")
```

```
Text(0, 0.5, 'Attempt rate')
```

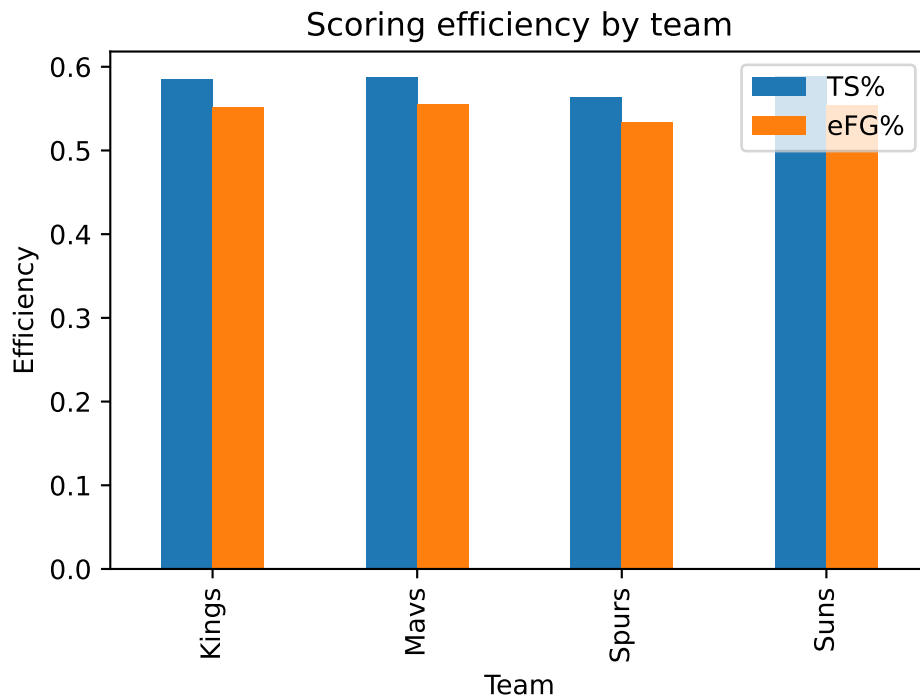
```
<Figure size 1650x1050 with 0 Axes>
```



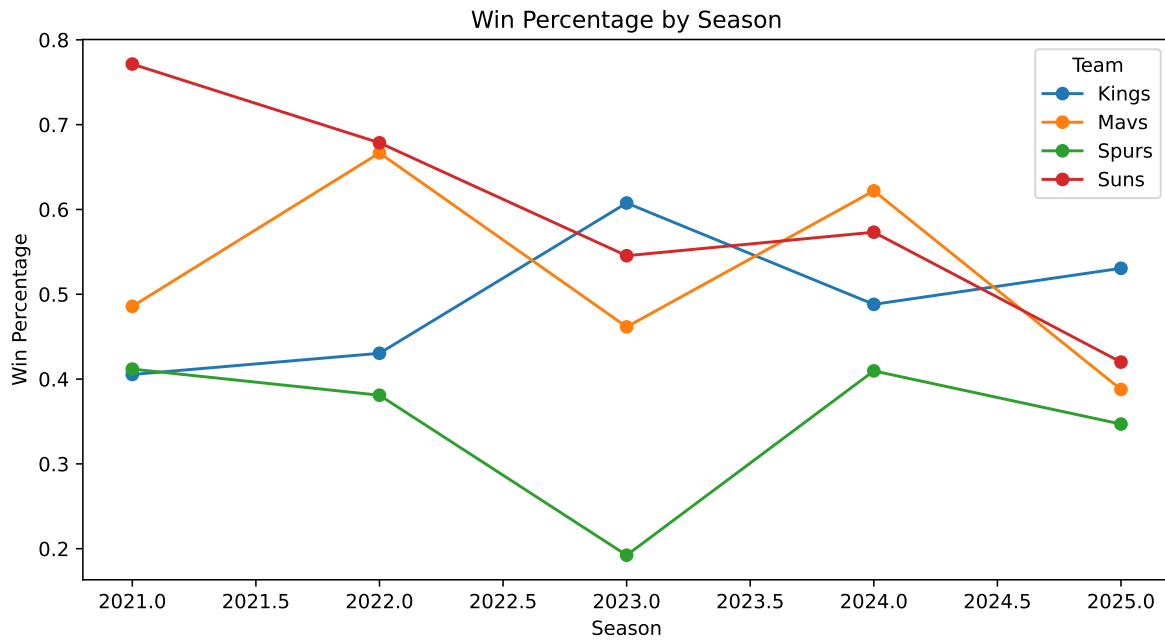
```
# Efficiency by Team
plt.figure()
efficiency = eda_df.groupby("Team")[["TS%", "eFG%"]].mean()
efficiency.plot(kind = "bar")
plt.ylabel("Efficiency")
plt.title("Scoring efficiency by team")
```

Text(0.5, 1.0, 'Scoring efficiency by team')

<Figure size 1650x1050 with 0 Axes>



```
# Win percentage by team and season
season_team = eda_df.groupby(["Season", "Team"])["Win"].mean().unstack()
season_team.plot(figsize=(10,5), marker="o")
plt.title("Win Percentage by Season")
plt.xlabel("Season")
plt.ylabel("Win Percentage")
plt.show()
```



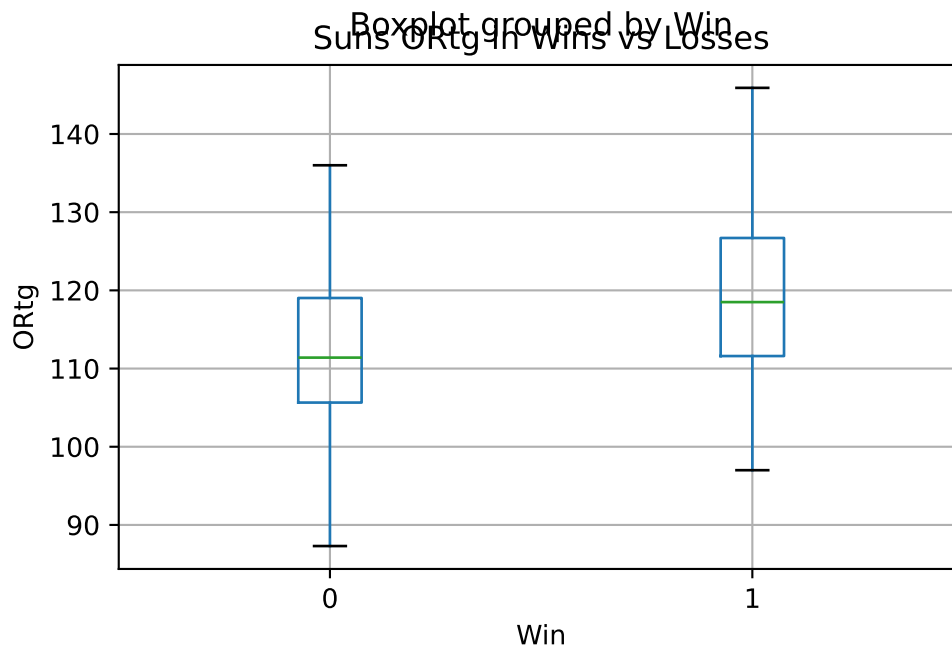
### Suns-only

```
# Offensive rating in wins vs losses
plt.figure()
suns_df.boxplot(column = "ORtg", by = "Win")
plt.title("Suns ORtg in Wins vs Losses")
plt.xlabel("Win")
plt.ylabel("ORtg")
```

Text(0, 0.5, 'ORtg')

<Figure size 1650x1050 with 0 Axes>

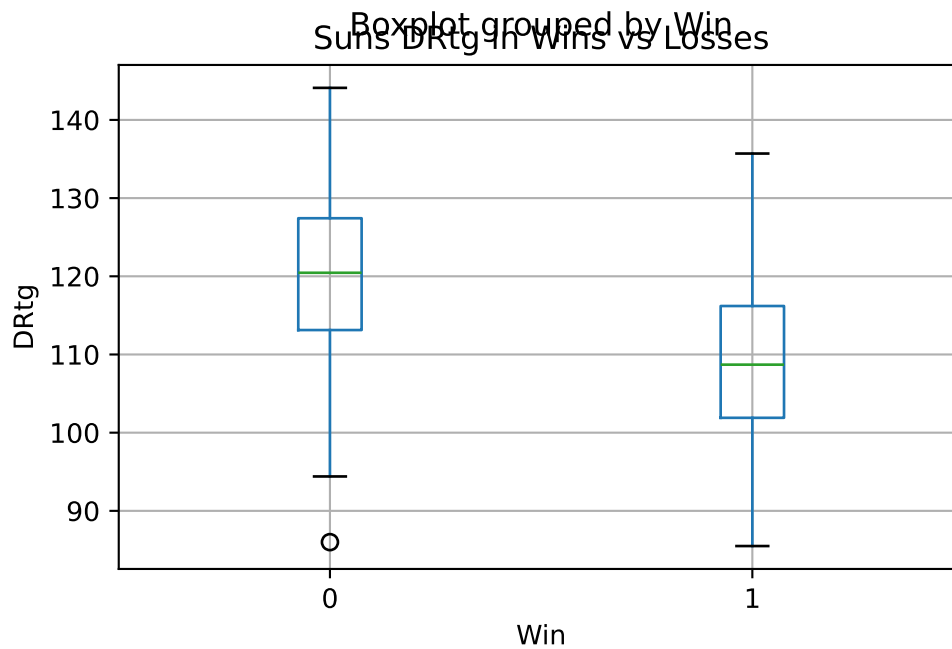




```
# Defensive rating in wins vs losses
plt.figure()
suns_df.boxplot(column = "DRtg", by = "Win")
plt.title("Suns DRtg in Wins vs Losses")
plt.xlabel("Win")
plt.ylabel("DRtg")
```

```
Text(0, 0.5, 'DRtg')
```

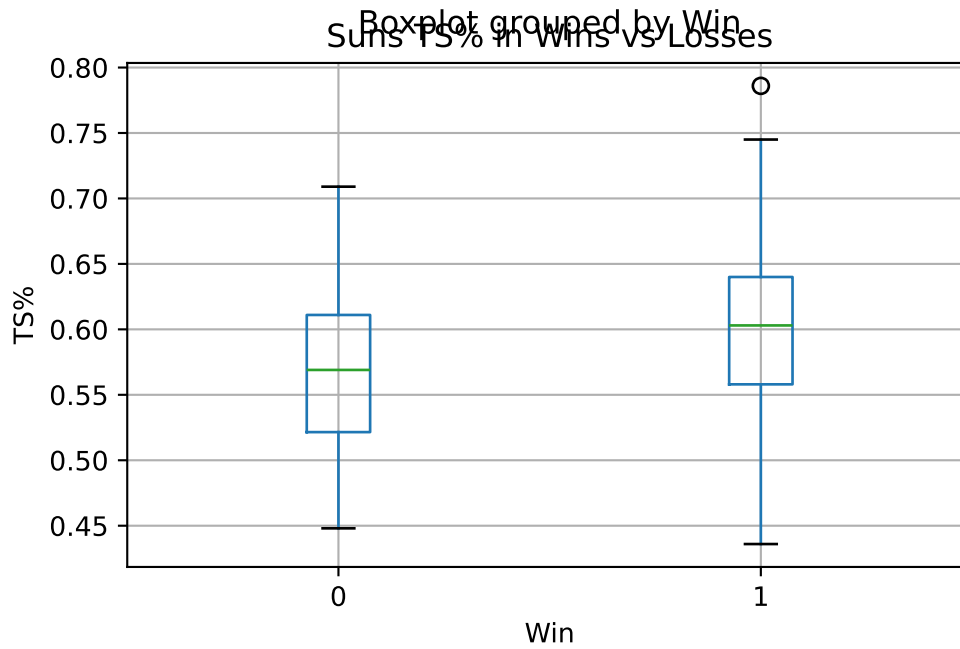
<Figure size 1650x1050 with 0 Axes>



```
# Suns TS% by win vs loss
plt.figure()
suns_df.boxplot(column = "TS%", by = "Win")
plt.title("Suns TS% in Wins vs Losses")
plt.xlabel("Win")
plt.ylabel("TS%")
```

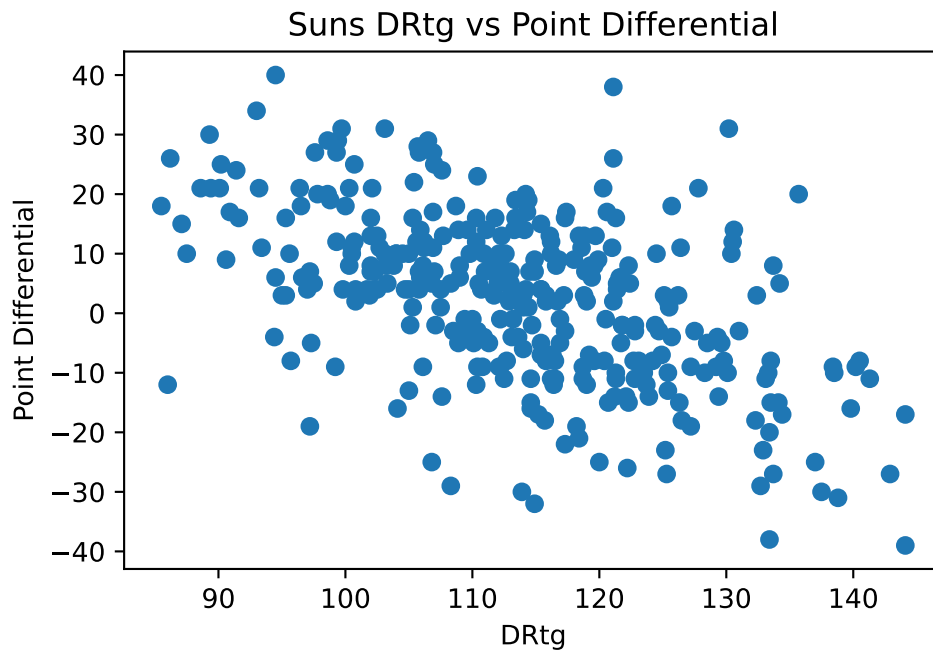
```
Text(0, 0.5, 'TS%')
```

```
<Figure size 1650x1050 with 0 Axes>
```



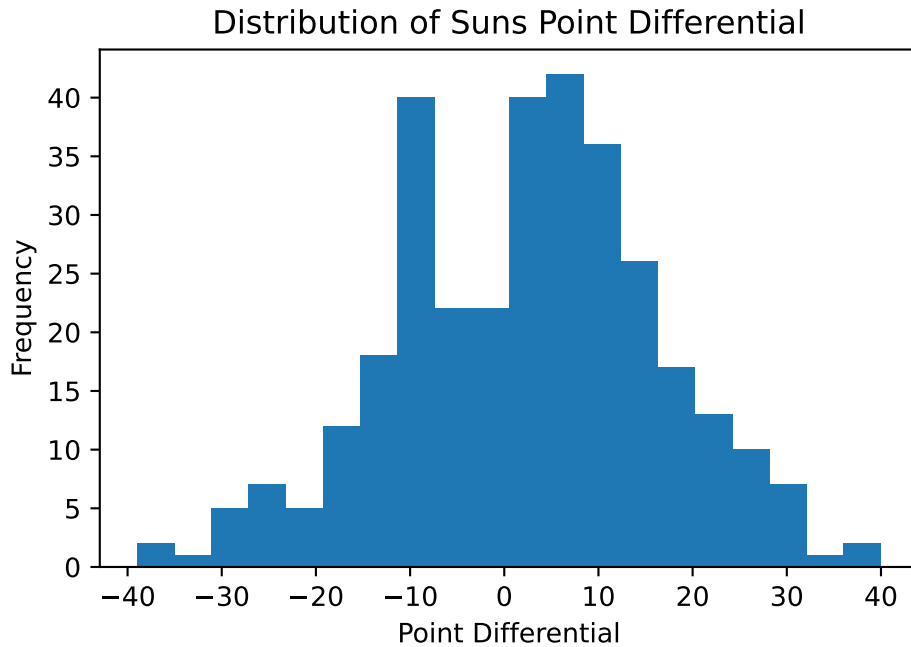
```
# Suns DRtg vs Point Differential
plt.figure()
plt.scatter(suns_df["DRtg"], suns_df["PointDiff"])
plt.title("Suns DRtg vs Point Differential")
plt.xlabel("DRtg")
plt.ylabel("Point Differential")
```

```
Text(0, 0.5, 'Point Differential')
```



```
plt.figure()
plt.hist(suns_df["PointDiff"], bins = 20)
plt.title("Distribution of Suns Point Differential")
plt.xlabel("Point Differential")
plt.ylabel("Frequency")
```

```
Text(0, 0.5, 'Frequency')
```



### EDA Main Conclusions

- Defense seems to be the strongest separator
  - Suns have highest win rate, best point differential, and best defensive rating
  - Suns and Mavs have similar offensive ratings, but the Suns perform better
  - The gap is better defense not more offense
- Efficiency matters more than 3 pointers
  - Suns do not have highest 3PAr
  - Mavs shoot more 3's but Suns win more
  - Suns have highest TS% and almost highest eFG% (wins/losses table)
  - TS% and eFG% are higher in Suns wins than losses
  - shot efficiency matter more than just taking more 3's
- Turnovers matter, but possibly less than defense & efficiency
  - Suns turnovers are lower in wins than losses
- Offensive rebounding probably not important for Suns
  - ORB% barely changes between Suns wins and losses
- 3 point attempt rate is prob going to be a weak predictor

- Mavs have highest 3PAr
- 3PAr is a little lower in wins than losses for Suns

## EDA Takeaways for Regression

- DRtg should be one of the main predictors (lower DRtg should correlate with more winning)
- include either TS% or eFG% at first, not both at the same time since they are closely related
- ORB% probably won't be significant
- 3PAr might be significant, but small effect
- Suns winning is driven more by defensive performance and scoring efficiency than by shot selection
- All teams regression:
  - $\text{Win} \sim \text{ORtg} + \text{DRtg} + 3\text{PAr} + \text{TOV\%} + \text{Home}$  (logistic)
  - $\text{PointDiff} \sim \text{ORtg} + \text{DRtg} + 3\text{PAr} + \text{TOV\%} + \text{Home}$  (linear)
  - answer how ORtg, DRtg, and shot selection relate to winning & how Suns compare to other teams
- Suns-Only Regression
  - $\text{Win} \sim \text{ORtg} + \text{DRtg} + 3\text{PAr} + \text{TOV\%} + \text{Home}$  (logistic)
  - $\text{PointDiff} \sim \text{ORtg} + \text{DRtg} + 3\text{PAr} + \text{TOV\%} + \text{Home}$  (linear)
  - answer what predicts Suns wins specifically & what matters most for Suns game to game

How do offensive efficiency, defensive efficiency, and shot selection influence the Phoenix Suns' probability of winning games compared to top NBA teams?

## Combined Teams - Linear Regression

```
# Combined teams regression
import statsmodels.api as sm
reg_df = eda_df[["PointDiff", "ORtg", "DRtg", "3PAr", "TOV%", "Home"]].dropna()
X = reg_df[["ORtg", "DRtg", "3PAr", "TOV%", "Home"]]
y = reg_df["PointDiff"]
X = sm.add_constant(X)
model_linear = sm.OLS(y, X).fit()
print(model_linear.summary())
```

# OLS Regression Results

|                   |                  |                     |           |       |        |        |
|-------------------|------------------|---------------------|-----------|-------|--------|--------|
| Dep. Variable:    | PointDiff        | R-squared:          | 0.887     |       |        |        |
| Model:            | OLS              | Adj. R-squared:     | 0.887     |       |        |        |
| Method:           | Least Squares    | F-statistic:        | 2057.     |       |        |        |
| Date:             | Thu, 30 Apr 2026 | Prob (F-statistic): | 0.00      |       |        |        |
| Time:             | 11:18:29         | Log-Likelihood:     | -         |       |        |        |
| 4007.5            |                  |                     |           |       |        |        |
| No. Observations: | 1311             | AIC:                | 8027.     |       |        |        |
| Df Residuals:     | 1305             | BIC:                | 8058.     |       |        |        |
| Df Model:         | 5                |                     |           |       |        |        |
| Covariance Type:  | nonrobust        |                     |           |       |        |        |
| =====             |                  |                     |           |       |        |        |
|                   | coef             | std err             | t         | P> t  | [0.025 | 0.975] |
| -----             |                  |                     |           |       |        |        |
| const             | 0.8498           | 2.161               | 0.393     | 0.694 | -3.391 | 5.090  |
| ORtg              | 0.9358           | 0.013               | 73.180    | 0.000 | 0.911  | 0.961  |
| DRtg              | -0.9391          | 0.012               | -77.059   | 0.000 | -0.963 | -      |
| 0.915             |                  |                     |           |       |        |        |
| 3PAr              | -0.0425          | 1.937               | -0.022    | 0.982 | -3.842 | 3.757  |
| TOV%              | -0.0416          | 0.045               | -0.918    | 0.359 | -0.131 | 0.047  |
| Home              | 0.0520           | 0.287               | 0.181     | 0.857 | -0.512 | 0.616  |
| =====             |                  |                     |           |       |        |        |
| Omnibus:          | 584.115          | Durbin-Watson:      | 3.010     |       |        |        |
| Prob(Omnibus):    | 0.000            | Jarque-Bera (JB):   | 73173.921 |       |        |        |
| Skew:             | -1.036           | Prob(JB):           | 0.00      |       |        |        |
| Kurtosis:         | 39.541           | Cond. No.           | 2.68e+03  |       |        |        |
| =====             |                  |                     |           |       |        |        |

## Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 2.68e+03. This might indicate that there are strong multicollinearity or other numerical problems.

- $R^2 = 0.887$
- $ORtg = 0.936$  with a  $p < 0.001$ , strong positive effect, for every +1 increase in offensive rating, point differential increases by 0.94 points
- $DRtg = -0.939$  with  $p < 0.001$ , strong negative effect, for every +1 increase in defensive rating (worse defense), point differential decreases by 0.94
- $ORtg$  and  $DRtg$  have almost identical magnitude but  $DRtg$  has a slightly larger absolute t value so defense is the strongest predictor
- $3PAr$ :  $p = 0.982$ , shooting more 3s doesn't meaningfully impact margin
- $TOV\%$ :  $p = 0.359$ , not statistically significant in this model

- Home:  $p = 0.857$ , no meaningful effect here

## Suns Only - Linear Regression

```
# Suns only regression
suns_reg = suns_df[["PointDiff", "ORtg", "DRtg", "3PAr", "TOV%", "Home"]].dropna()
X_suns = suns_reg[["ORtg", "DRtg", "3PAr", "TOV%", "Home"]]
y_suns = suns_reg["PointDiff"]
X_suns = sm.add_constant(X_suns)
model_suns_linear = sm.OLS(y_suns, X_suns).fit()
print(model_suns_linear.summary())
```

```

                                OLS Regression Results
=====
Dep. Variable:                  PointDiff    R-squared:                  0.547
Model:                            OLS      Adj. R-squared:              0.540
Method:                    Least Squares    F-statistic:                  77.66
Date:                Thu, 30 Apr 2026      Prob (F-statistic):          3.42e-
53
Time:                        11:18:29      Log-Likelihood:              -
1202.3
No. Observations:                327      AIC:                        2417.
Df Residuals:                    321      BIC:                        2439.
Df Model:                        5
Covariance Type:                nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const          13.4107         7.689         1.744      0.082      -1.717      28.539
ORtg           0.6706         0.053        12.740      0.000         0.567         0.774
DRtg          -0.7433         0.046       -16.210      0.000        -0.834         -
0.653
3PAr          -2.7568         7.331        -0.376      0.707       -17.179      11.665
TOV%          -0.2775         0.170        -1.631      0.104        -0.612         0.057
Home           0.4081         1.083         0.377      0.707        -1.723         2.539
=====
Omnibus:                    58.841    Durbin-Watson:              2.877
Prob(Omnibus):              0.000    Jarque-Bera (JB):          432.216
Skew:                      -0.471    Prob(JB):                  1.40e-
94

```



Kurtosis: 8.553 Cond. No. 2.47e+03  
=====

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.  
[2] The condition number is large, 2.47e+03. This might indicate that there are strong multicollinearity or other numerical problems.

- $R^2 = 0.547$ , lower than combined team model so game-to-game variability matters more for one team
- $ORtg = 0.671$  with a  $p < 0.001$ , strong positive effect
- $DRtg = -0.743$  with a  $p < 0.001$ , strong negative effect
- Both offensive and defensive efficiency significantly influence performance, with defensive rating again showing a slightly stronger effect. Shot selection remains insignificant, showing that the Suns' success is driven more by efficiency and defense than by increasing three-point volume

## Combined Teams - Logistic Regression

```
import statsmodels.formula.api as smf
# Create "Win" column from "PointDiff"
reg_df["Win"] = (reg_df["PointDiff"] > 0).astype(int)

# Combined teams logistic regression
log_fit_combined = smf.logit("Win ~ ORtg + DRtg + Q('3PAr') + Q('TOV%') + Home", data=reg_df)
print(log_fit_combined.summary())
```

Optimization terminated successfully.

Current function value: 0.219328

Iterations 9

### Logit Regression Results

```
=====
Dep. Variable:          Win    No. Observations:          1311
Model:                Logit    Df Residuals:              1305
Method:                MLE     Df Model:                  5
Date:                 Thu, 30 Apr 2026    Pseudo R-squ.:        0.6835
Time:                 11:18:29    Log-Likelihood:        -
287.54
converged:              True    LL-Null:              -
908.63
```

Covariance Type: nonrobust LLR p-value: 2.142e-266

|           | coef    | std err | z       | P> z  | [0.025 | 0.975] |
|-----------|---------|---------|---------|-------|--------|--------|
| Intercept | -0.6676 | 1.685   | -0.396  | 0.692 | -3.970 | 2.635  |
| ORtg      | 0.3629  | 0.023   | 15.981  | 0.000 | 0.318  | 0.407  |
| DRtg      | -0.3559 | 0.022   | -16.006 | 0.000 | -0.399 | -      |
| 0.312     |         |         |         |       |        |        |
| Q('3PAr') | -1.0287 | 1.420   | -0.725  | 0.469 | -3.811 | 1.754  |
| Q('TOV%') | 0.0223  | 0.035   | 0.642   | 0.521 | -0.046 | 0.090  |
| Home      | -0.0190 | 0.215   | -0.088  | 0.930 | -0.440 | 0.402  |

- $R^2 = 0.684$ , explaining a lot of the variation in wins/losses
- ORtg = 0.363 with a  $p < 0.001$ , statistically significant  $\rightarrow$  higher means increasing win probability
- DRtg = -0.356 with a  $p < 0.001$ , statistically significant  $\rightarrow$  lower DRtg means better defense
- ORtg & DRtg are nearly equal in magnitude, so offense and defense contribute equally to winning
- 3PAr:  $p = 0.469$ , not significant
- TOV%:  $p = 0.521$ , not significant
- Home:  $p = 0.930$ , not significant

## Suns Only - Logistic Regression

```
import statsmodels.formula.api as smf
# Suns only logistic regression
log_fit = smf.logit("Win ~ ORtg + DRtg + Q('3PAr') + Q('TOV%') + Home", data=suns_df).fit()
print(log_fit.summary())
```

Optimization terminated successfully.

Current function value: 0.418369

Iterations 7

### Logit Regression Results

|                |                  |                   |        |
|----------------|------------------|-------------------|--------|
| Dep. Variable: | Win              | No. Observations: | 327    |
| Model:         | Logit            | Df Residuals:     | 321    |
| Method:        | MLE              | Df Model:         | 5      |
| Date:          | Thu, 30 Apr 2026 | Pseudo R-squ.:    | 0.3818 |

Time: 11:18:29 Log-Likelihood: -  
 136.81  
 converged: True LL-Null: -  
 221.31  
 Covariance Type: nonrobust LLR p-value: 1.192e-34

|           | coef    | std err | z      | P> z  | [0.025 | 0.975] |
|-----------|---------|---------|--------|-------|--------|--------|
| Intercept | 4.1163  | 2.214   | 1.860  | 0.063 | -0.222 | 8.455  |
| ORtg      | 0.1330  | 0.019   | 7.131  | 0.000 | 0.096  | 0.170  |
| DRtg      | -0.1538 | 0.019   | -8.267 | 0.000 | -0.190 | -      |
| 0.117     |         |         |        |       |        |        |
| Q('3PAR') | -1.5102 | 2.062   | -0.732 | 0.464 | -5.552 | 2.532  |
| Q('TOV%') | -0.0847 | 0.049   | -1.731 | 0.084 | -0.181 | 0.011  |
| Home      | 0.3128  | 0.304   | 1.029  | 0.304 | -0.283 | 0.909  |

- $R^2 = 0.382$ , weaker fit than combined teams model, so there are most likely other factors (not included) influencing the Suns wins
- $ORtg = 0.133$  with a  $p < 0.001$ , statistically significant  $\rightarrow$  higher means increasing win probability
- $DRtg = -0.154$  with a  $p < 0.001$ , statistically significant  $\rightarrow$  lower  $DRtg$  means better defense, increasing win probability
- $DRtg$  has a slightly stronger effect than  $ORtg$  for Suns.
- 3PAR:  $p = 0.464$ , not significant
- TOV%:  $p = 0.084$ , not significant at the 0.05 level

## Validations (CI, Hypothesis Testing)

```
# Lasso Regression for entire data
from sklearn.linear_model import LassoCV
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
X = reg_df[["ORtg", "DRtg", "3PAR", "TOV%", "Home"]]
y = reg_df["PointDiff"]
lasso_pipeline = Pipeline([("scaler", StandardScaler()), ("lasso", LassoCV(cv = 5))])
lasso_pipeline.fit(X, y)
lasso = lasso_pipeline.named_steps["lasso"]
coefficients = pd.Series(lasso.coef_, index = X.columns)
print(coefficients)
```

```

ORtg    10.949180
DRtg    -11.038059
3PAr    -0.000000
TOV%    -0.060826
Home     0.000000
dtype: float64

```

- ORtg and DRtg are very dominant predictors with opposite signs
- TOV% had a tiny coefficient which indicates it has a very small effect on point differential
- 3PAr and Home court advantage shrunk towards zero and are not influential in predicting the point differential

```

# Lasso Regression for suns only
X_suns = suns_reg[["ORtg", "DRtg", "3PAr", "TOV%", "Home"]]
y_suns = suns_reg["PointDiff"]
lasso_pipeline.fit(X_suns, y_suns)
lasso_suns = lasso_pipeline.named_steps["lasso"]
coeff_suns = pd.Series(lasso_suns.coef_, index = X_suns.columns)
print(coeff_suns)

```

```

ORtg     7.087498
DRtg    -8.887560
3PAr    -0.058727
TOV%    -0.813990
Home     0.077879
dtype: float64

```

- Suns Lasso also had ORtg and DRtg has the most important predictors
- TOV% had a larger coefficient than the combined dataset and has a small effect on point differential
- 3PAr and Home court advantage were kept this time, but were very tiny
- TOV% is has a larger role than the entire dataset on point differential, but it is still mostly driven by ORtg and DRtg

```

# Confidence interval for linear model
ci_linear = model_linear.conf_int()
print(ci_linear)

```

```

          0          1
const -3.390535  5.090217
ORtg   0.910757  0.960932
DRtg  -0.963032 -0.915215

```

|      |           |          |
|------|-----------|----------|
| 3PAr | -3.841942 | 3.756916 |
| TOV% | -0.130531 | 0.047333 |
| Home | -0.511878 | 0.615781 |

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```
# Confidence interval for suns linear model
ci_suns_linear = model_suns_linear.conf_int()
print(ci_suns_linear)
```

|       | 0          | 1         |
|-------|------------|-----------|
| const | -1.717313  | 28.538631 |
| ORtg  | 0.567071   | 0.774194  |
| DRtg  | -0.833551  | -0.653118 |
| 3PAr  | -17.178743 | 11.665121 |
| TOV%  | -0.612110  | 0.057136  |
| Home  | -1.722893  | 2.539078  |

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```
# Confidence interval for logistic model
ci_log = log_fit_combined.conf_int()
print(ci_log)
```

|           | 0         | 1         |
|-----------|-----------|-----------|
| Intercept | -3.970364 | 2.635133  |
| ORtg      | 0.318389  | 0.407403  |
| DRtg      | -0.399462 | -0.312303 |
| Q('3PAr') | -3.811191 | 1.753768  |
| Q('TOV%') | -0.045784 | 0.090434  |
| Home      | -0.439830 | 0.401853  |

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```
# Confidence interval for suns logistic model
ci_suns_log = log_fit.conf_int()
print(ci_suns_log)
```

|           | 0         | 1         |
|-----------|-----------|-----------|
| Intercept | -0.222246 | 8.454876  |
| ORtg      | 0.096416  | 0.169503  |
| DRtg      | -0.190320 | -0.117372 |
| Q('3PAr') | -5.552466 | 2.532025  |
| Q('TOV%') | -0.180650 | 0.011228  |
| Home      | -0.283181 | 0.908687  |

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## Coaching Decisions

- Prioritize defensive performance because it has the biggest impact on winning margin
- Increasing 3-point attempts alone will not improve outcomes